Tema 3
Conductores en equilibrio electrostático

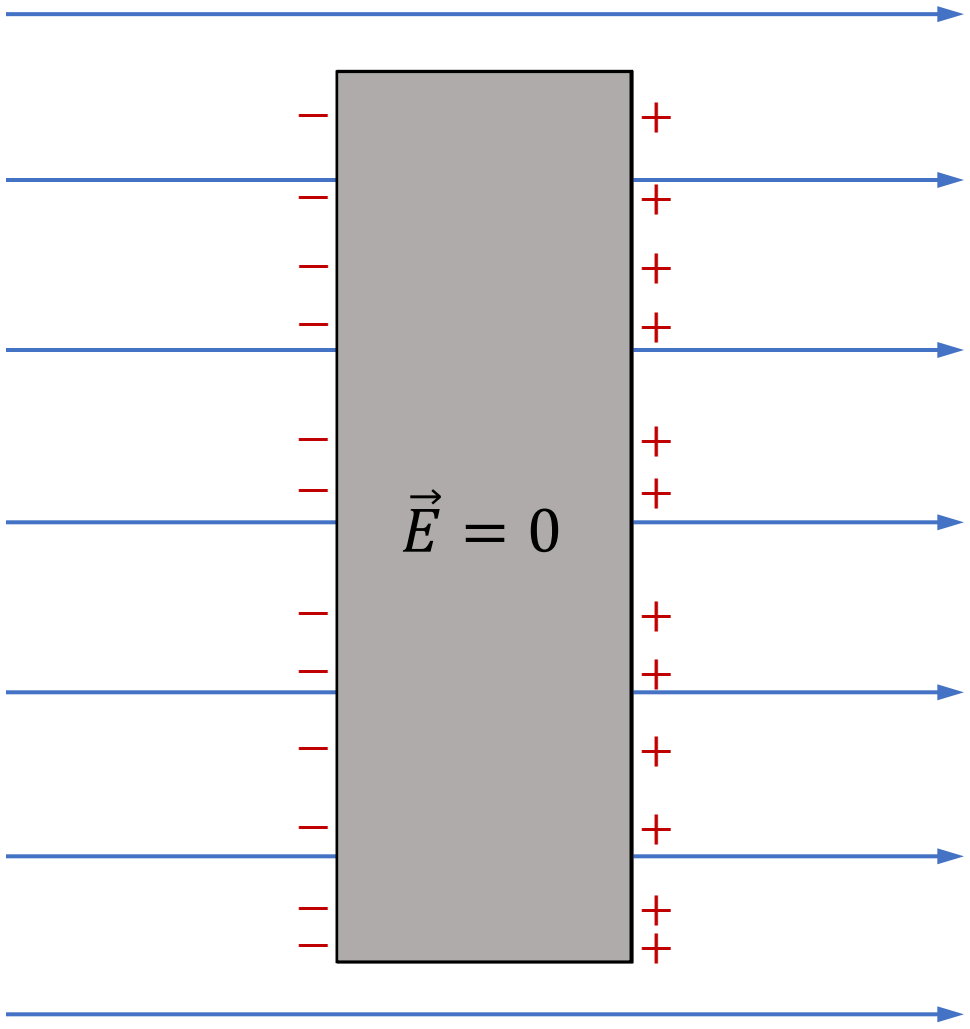
- Introducción
- Propiedades básicas
- Presión electrostática
- Cavidades en conductores
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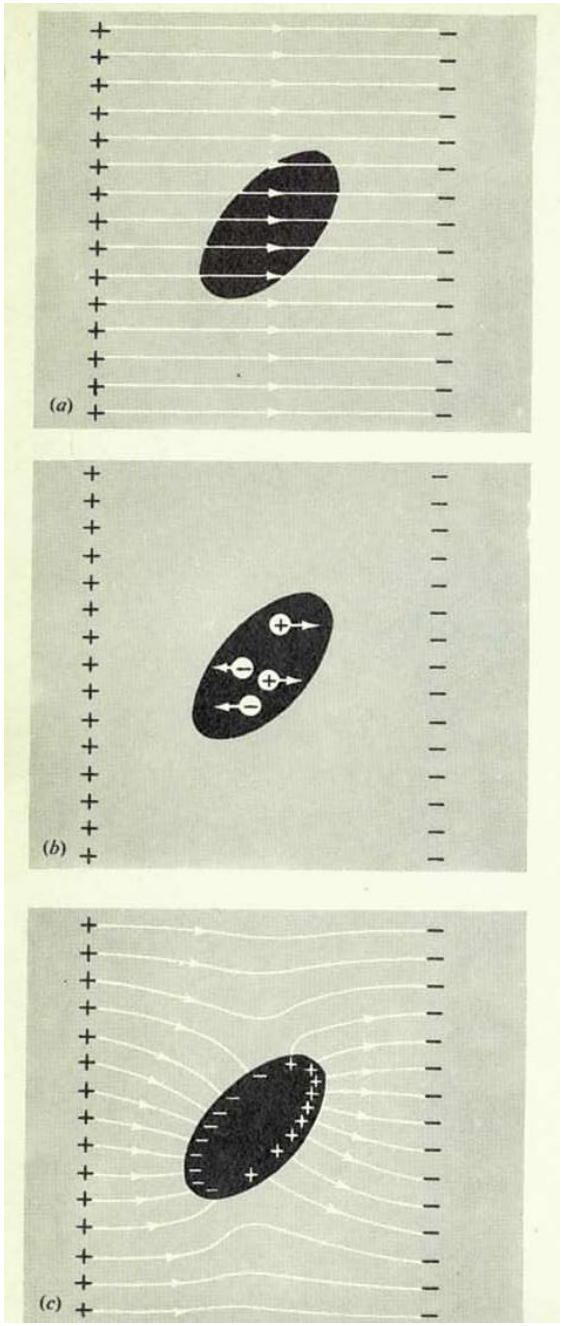
- **Introducción**
- **Propiedades básicas**
- Presión electrostática
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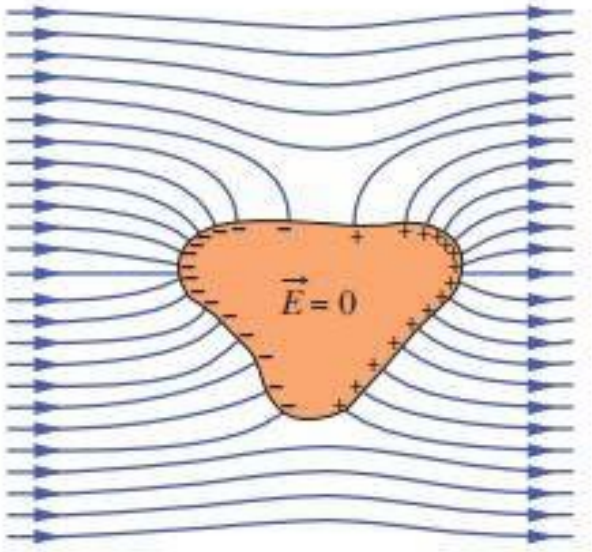
Introducción

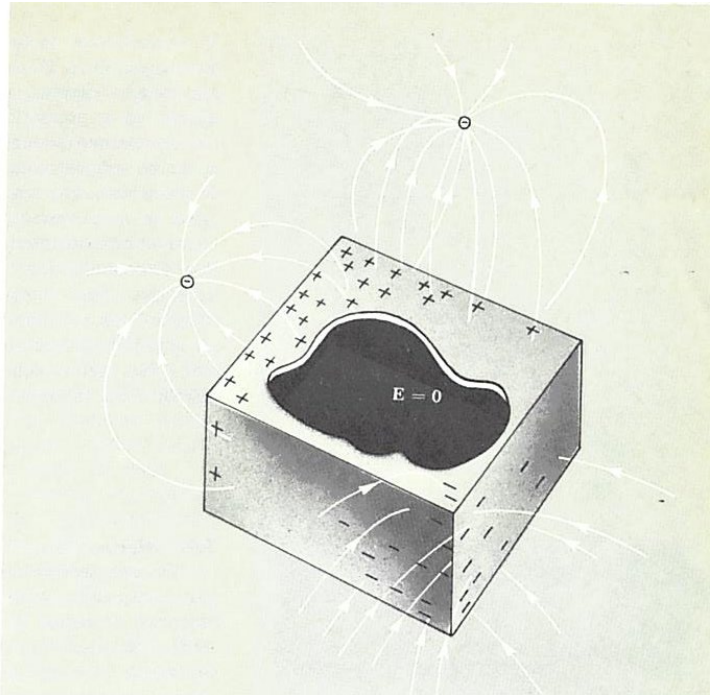
- ¿Qué es un **conductor**?
  - Las cargas se mueven libremente
  - Admite un número ilimitado de cargas
- En **equilibrio electrostático**
  - $\vec{E} = 0$
  - ¿Por qué ocurre esto?



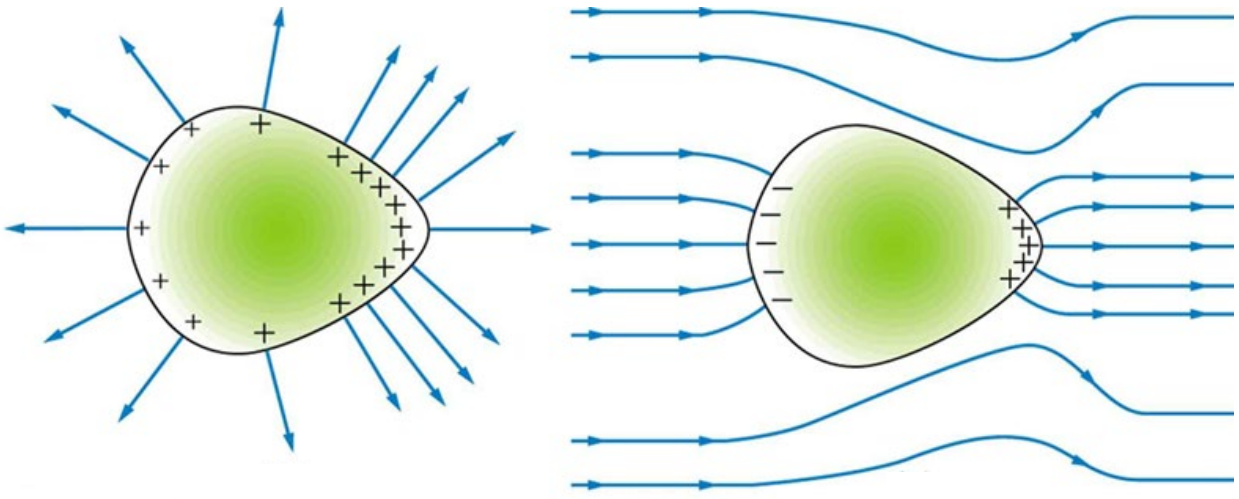
$\vec{E} = 0$ en el interior

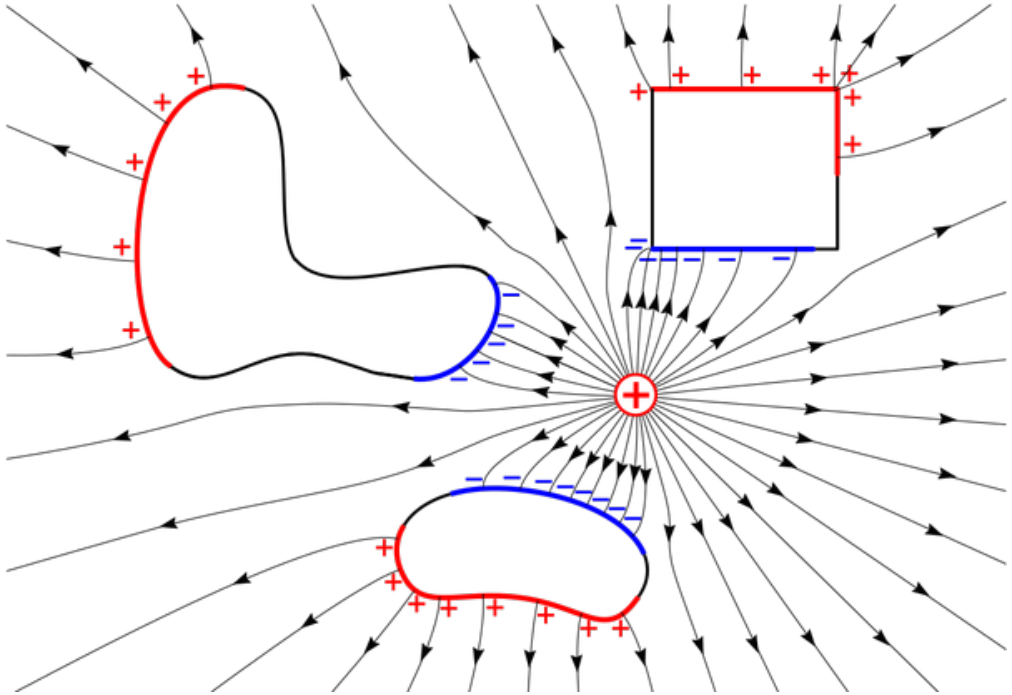






$\vec{E} = 0$ en el interior

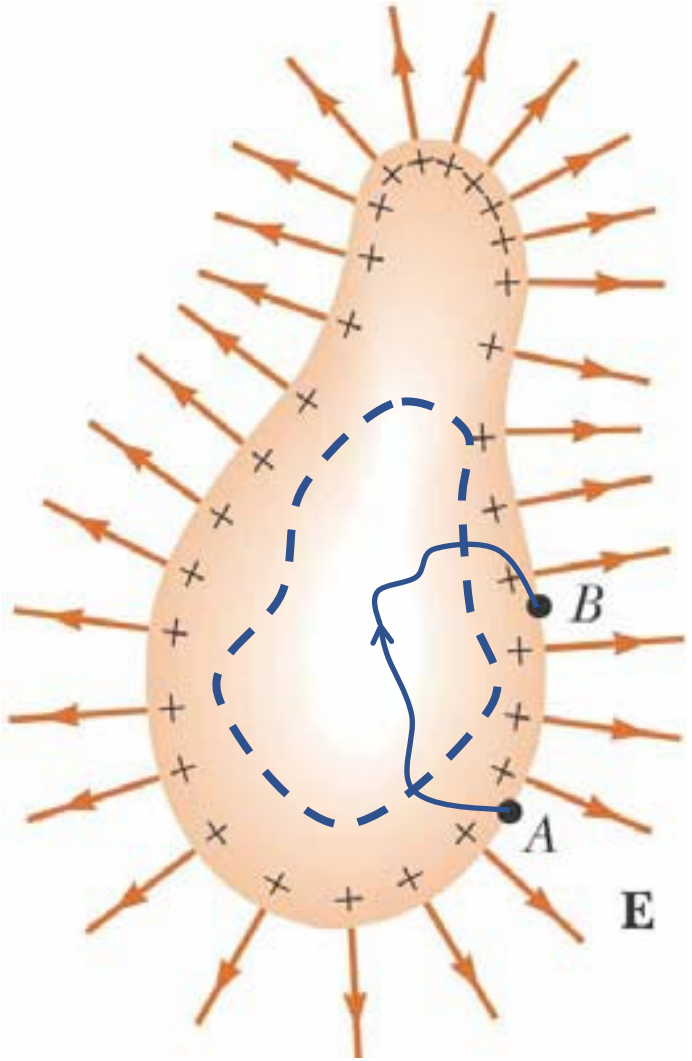




Propiedades básicas

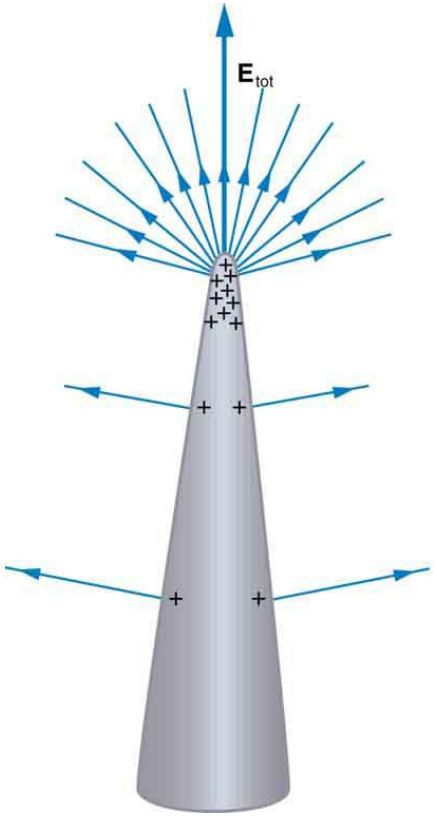
- Campo nulo en el interior ($\vec{E}(\vec{r}) = 0$)
- Potencial constante ($\phi(\vec{r}) = \text{cte}$)
- Carga en la superficie
- En la superficie el campo es normal

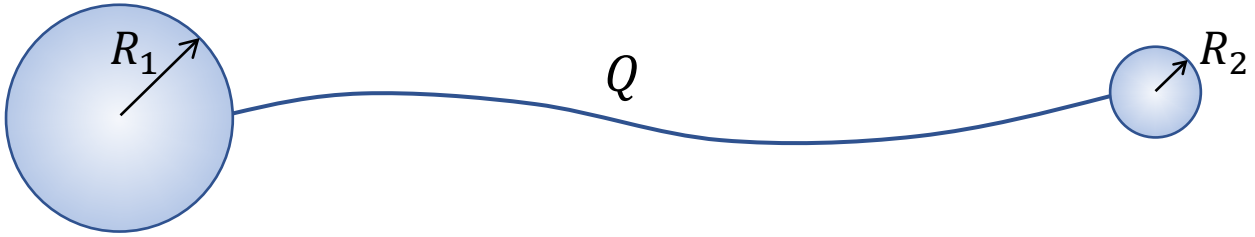
$$\vec{E} = \frac{\sigma}{\varepsilon_0}\hat{n}$$



$$\phi_B - \phi_A = -\int_A^B \vec{E} \cdot d\vec{\ell} = 0$$

Efecto de puntas

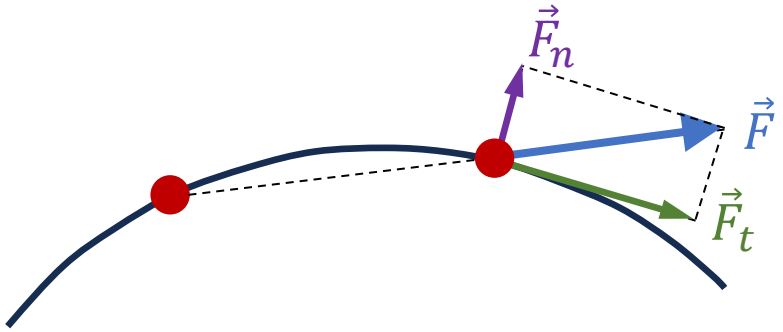


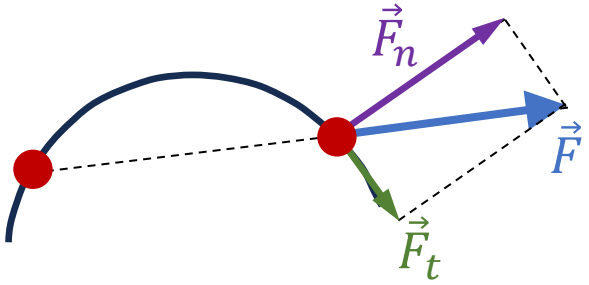


$$Q = Q_1 + Q_2$$

$$\frac{Q_1}{4\pi\varepsilon_0 R_1} = \frac{Q_2}{4\pi\varepsilon_0 R_2}$$

$$\sigma_1 R_1 = \sigma_2 R_2$$

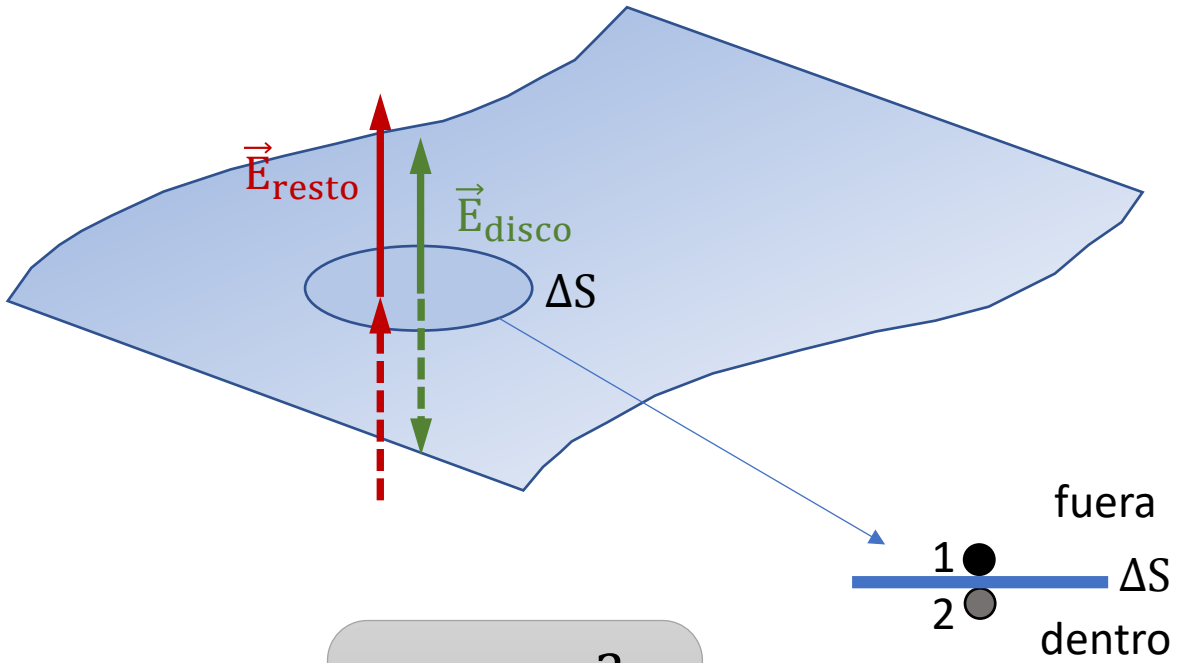




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Presión en la superficie de un conductor



$$P = \frac{\sigma^2}{2\varepsilon_0}$$

$$\vec{E} = \vec{E}_{disco} + \vec{E}_{resto}$$

$$\vec{E}_1 = \vec{E}_{resto} + \vec{E}_{disco,1} = \vec{E}_{resto} + \frac{\sigma}{2\varepsilon_0}\hat{n}$$

$$\vec{E}_2 = \vec{E}_{resto} + \vec{E}_{disco,2} = \vec{E}_{resto} - \frac{\sigma}{2\varepsilon_0}\hat{n}$$

$$\vec{E}_{resto} = \frac{1}{2}\left(\vec{E}_1 + \vec{E}_2\right) = \frac{\sigma}{2\varepsilon_0}$$

$$\Delta\vec{F} = \Delta q\ \vec{E}_{resto} = \Delta q \frac{\sigma}{2\varepsilon_0}$$

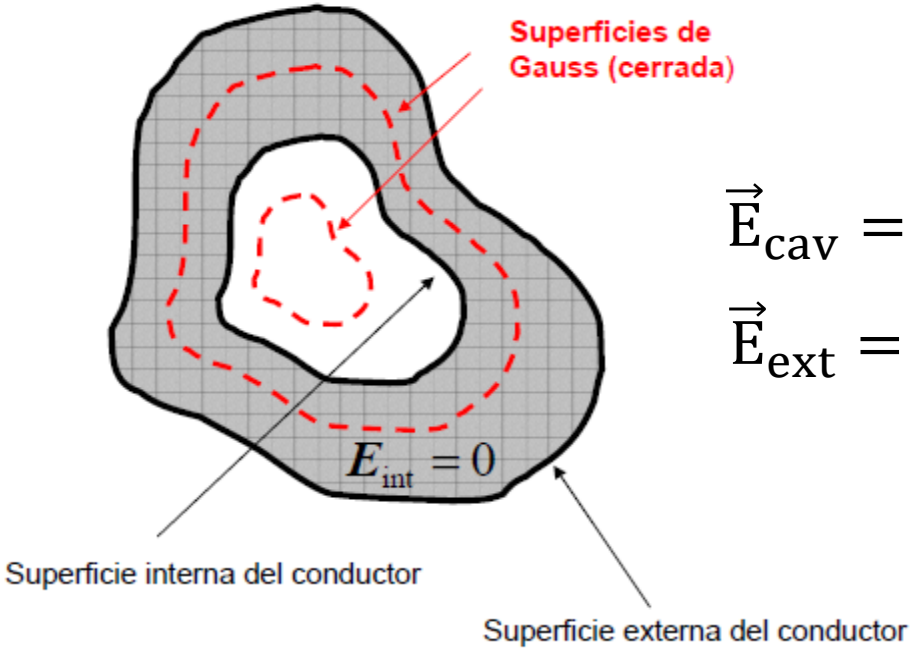
$$P = \frac{\Delta F}{\Delta S} = \frac{\Delta q}{\Delta S}\frac{\sigma}{2\varepsilon_0} = \frac{\sigma^2}{2\varepsilon_0}$$

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Cavidad vacía

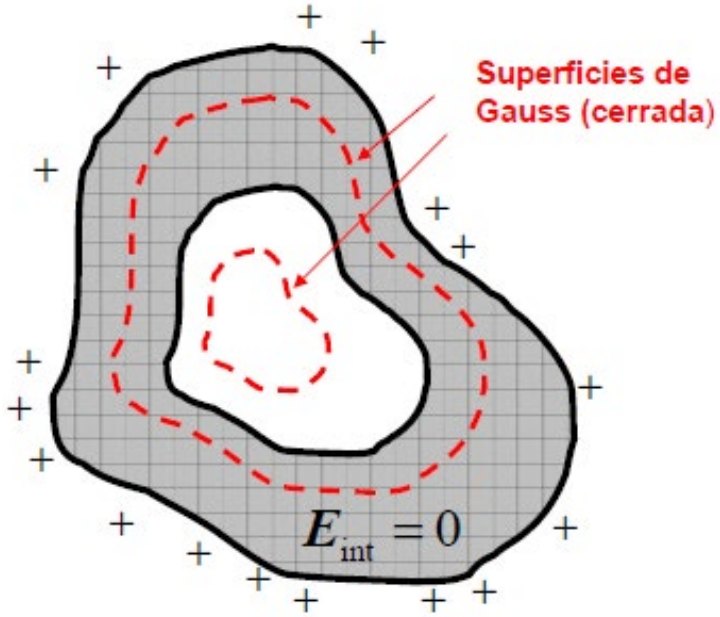
Conductor sin carga ($Q = 0$)



$$\vec{E}_{cav} = 0$$

$$\vec{E}_{ext} = 0$$

Conductor con carga ($Q \neq 0$)



$$Q_{sup.int} = 0$$

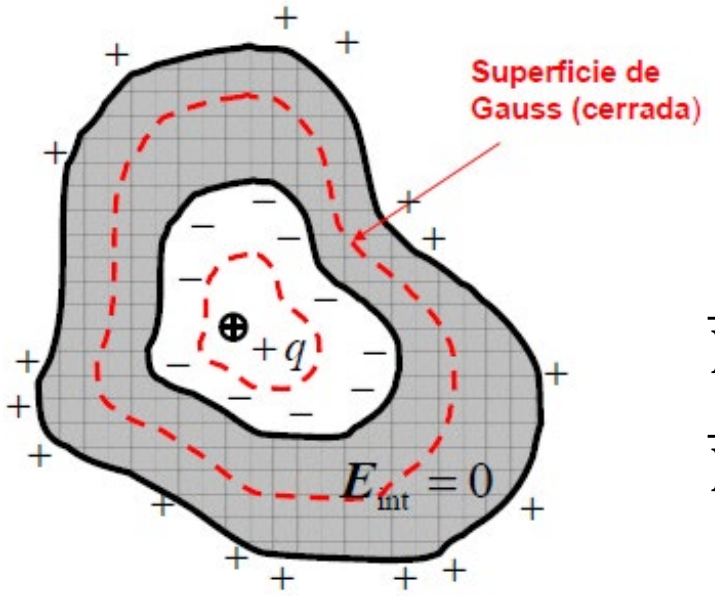
$$Q_{sup.ext} = Q$$

$$\vec{E}_{cav} = 0$$

$$\vec{E}_{ext} \neq 0$$

Cavidad con carga (q)

Conductor sin carga ($\mathrm{Q} = 0$)



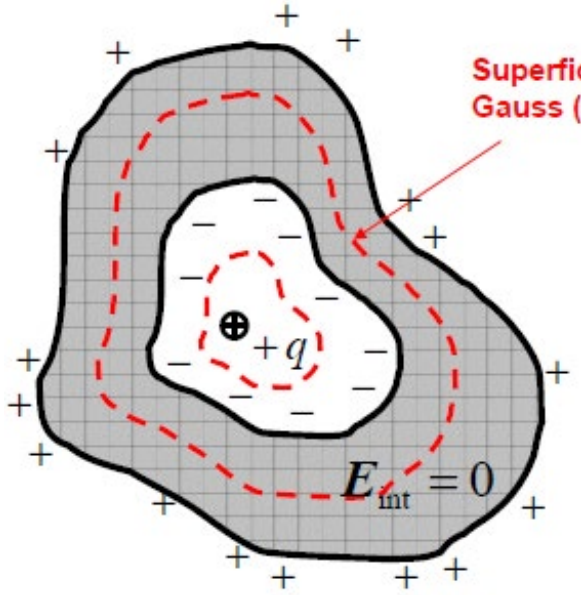
$$\mathrm{Q}_{\mathrm{sup.int}} = -q$$

$$\mathrm{Q}_{\mathrm{sup.ext}} = +q$$

$$\vec{\mathrm{E}}_{\mathrm{cav}} \neq 0$$

$$\vec{\mathrm{E}}_{\mathrm{ext}} \neq 0$$

Conductor con carga ($\mathrm{Q} \neq 0$)

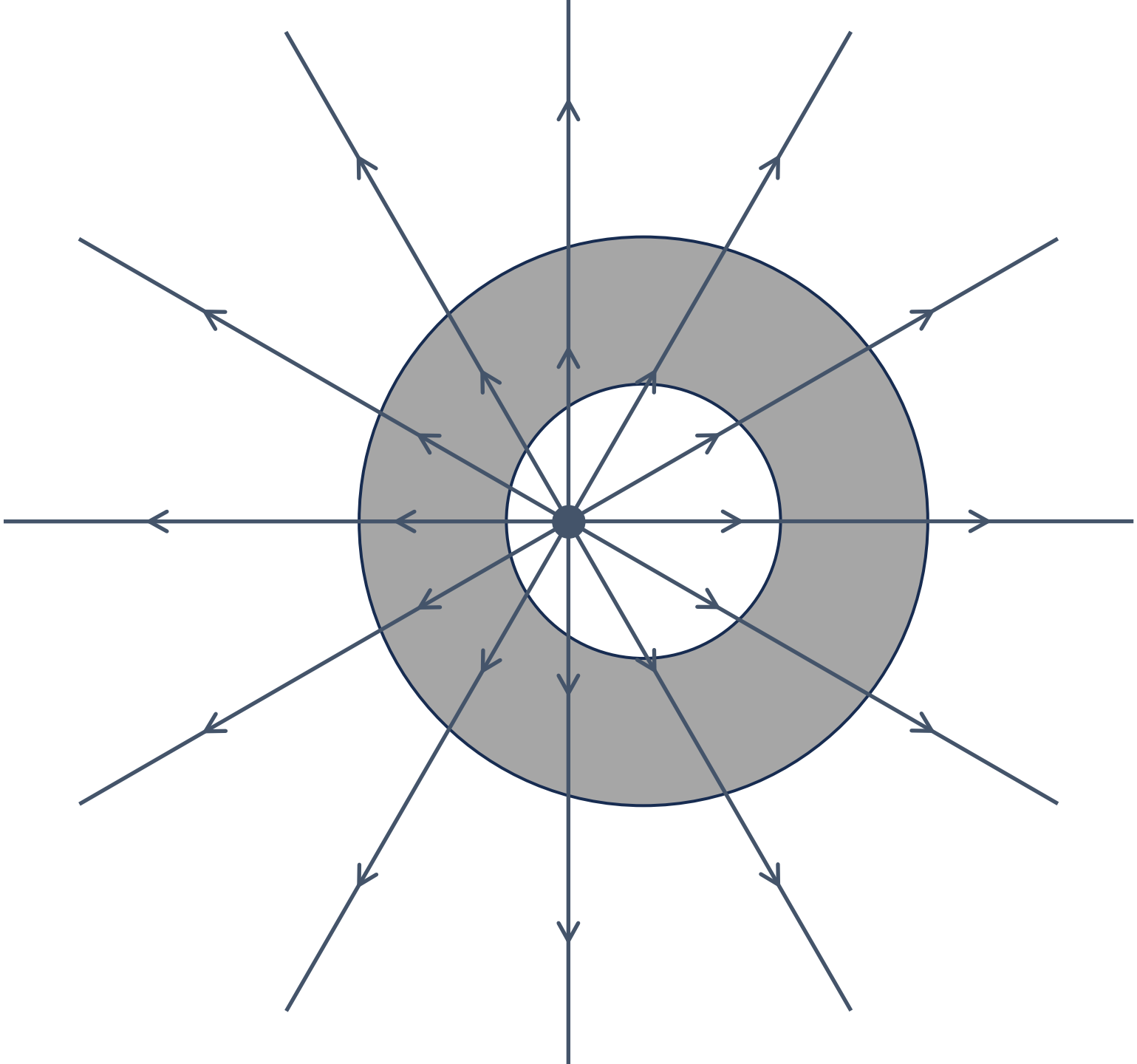


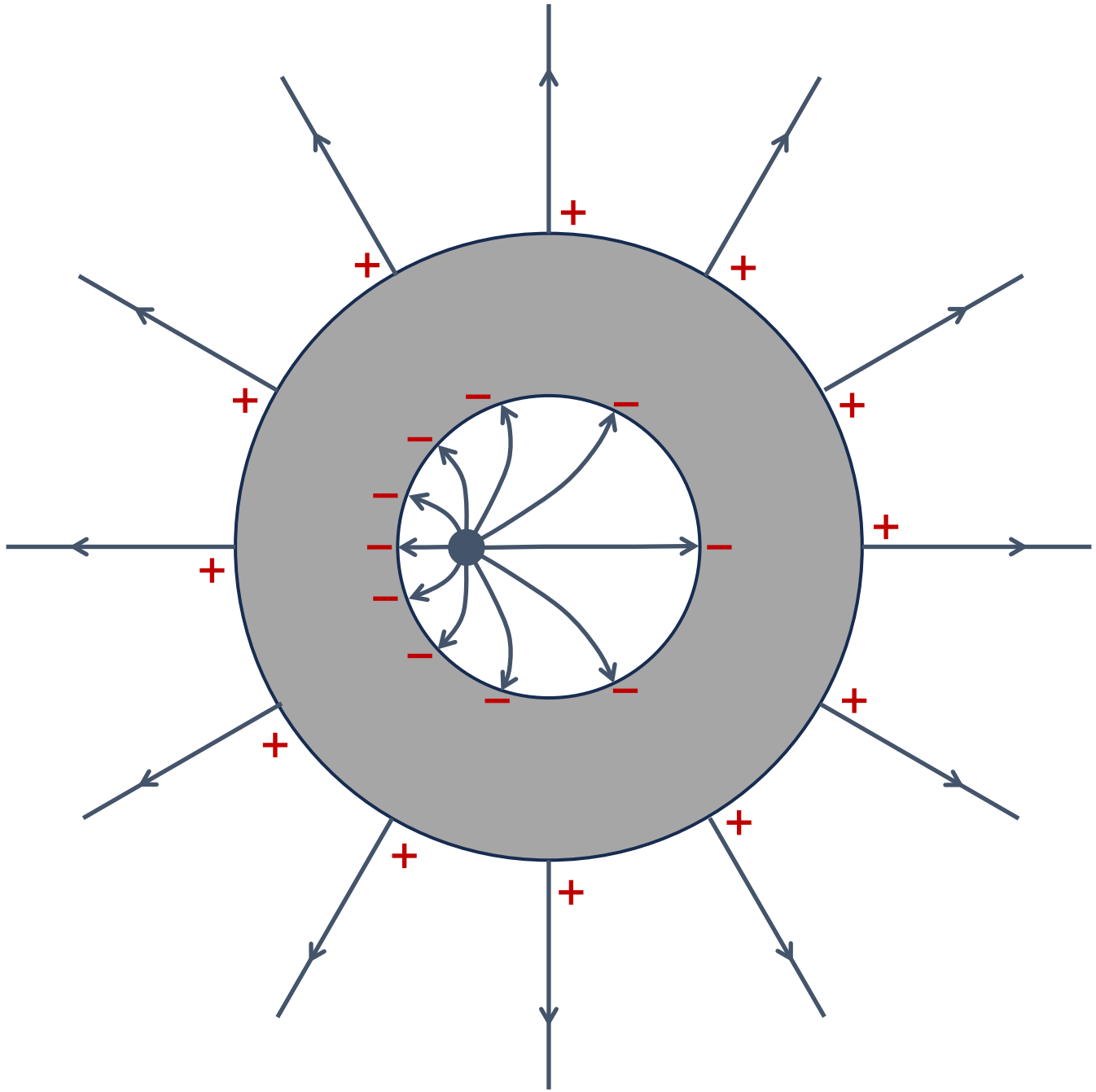
$$\mathrm{Q}_{\mathrm{sup.int}} = -q$$

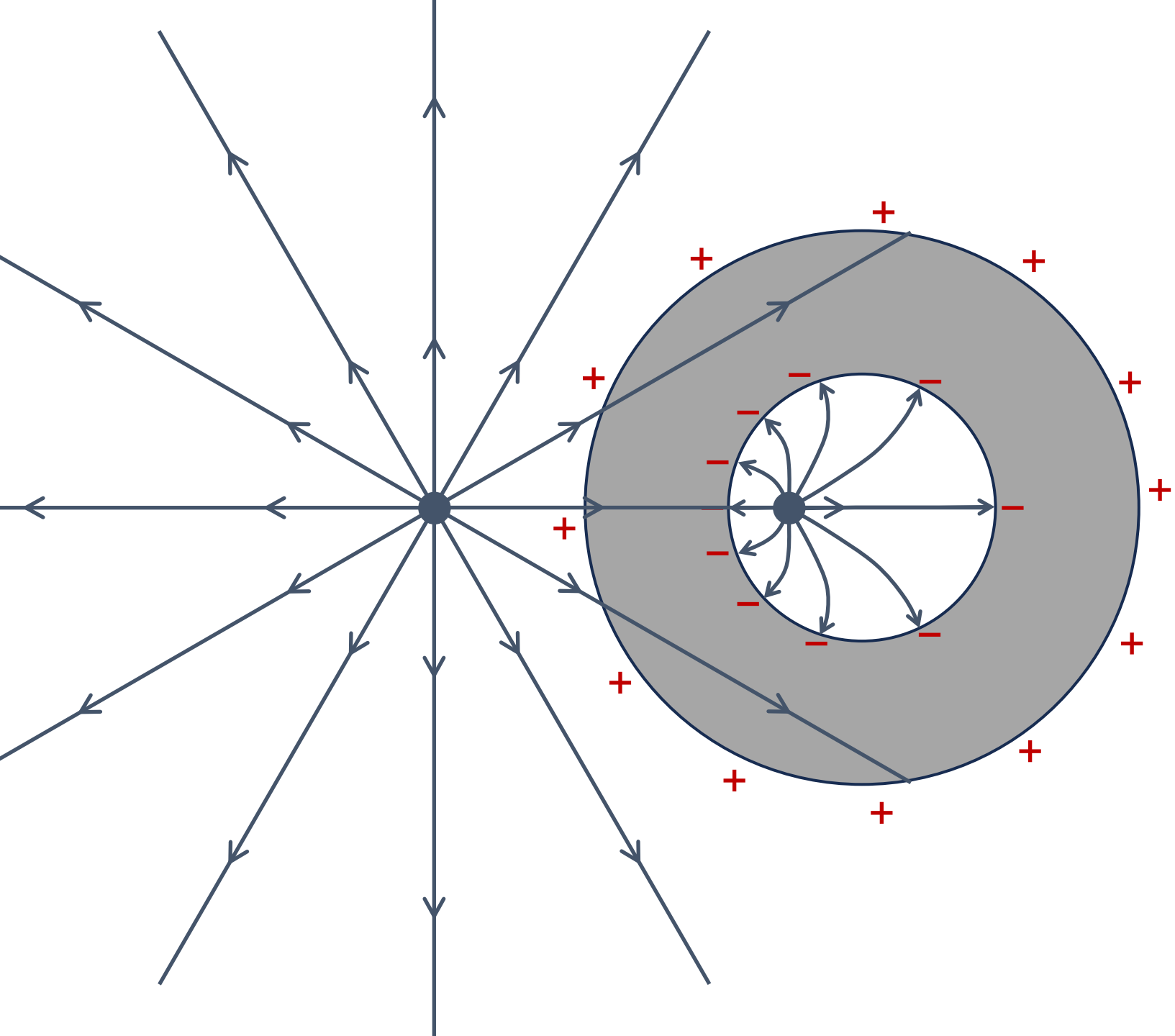
$$\mathrm{Q}_{\mathrm{sup.ext}} = Q + q$$

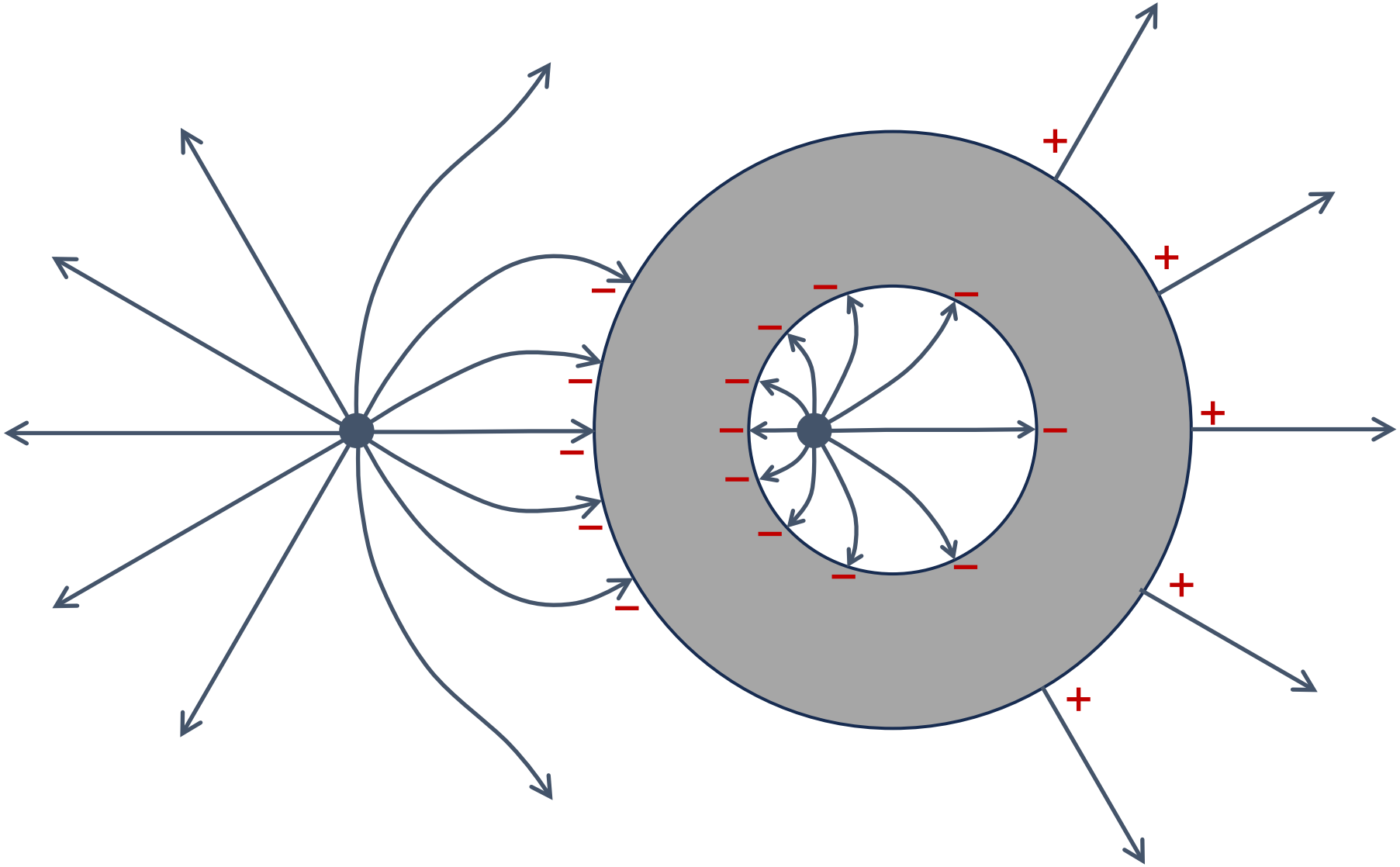
$$\vec{\mathrm{E}}_{\mathrm{cav}} \neq 0$$

$$\vec{\mathrm{E}}_{\mathrm{ext}} \neq 0$$



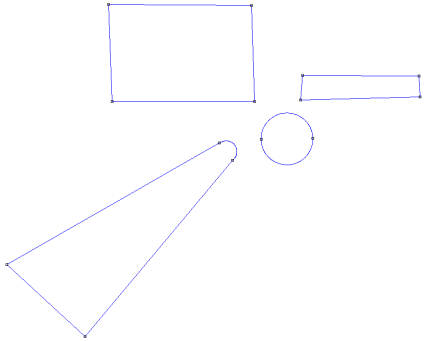


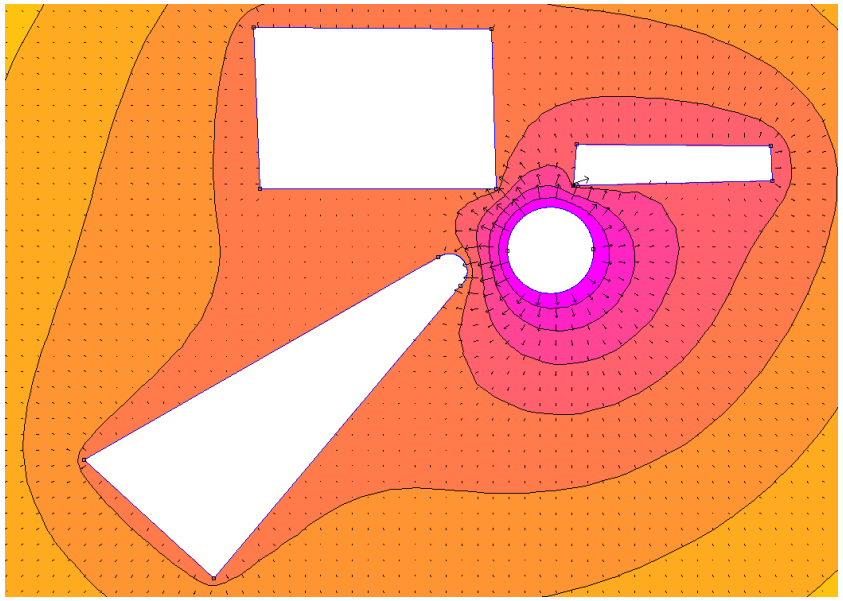


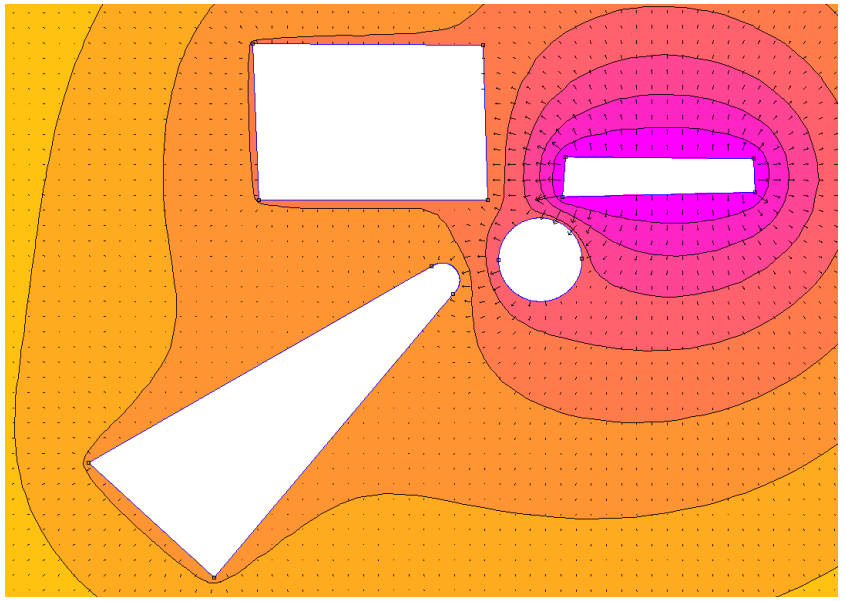


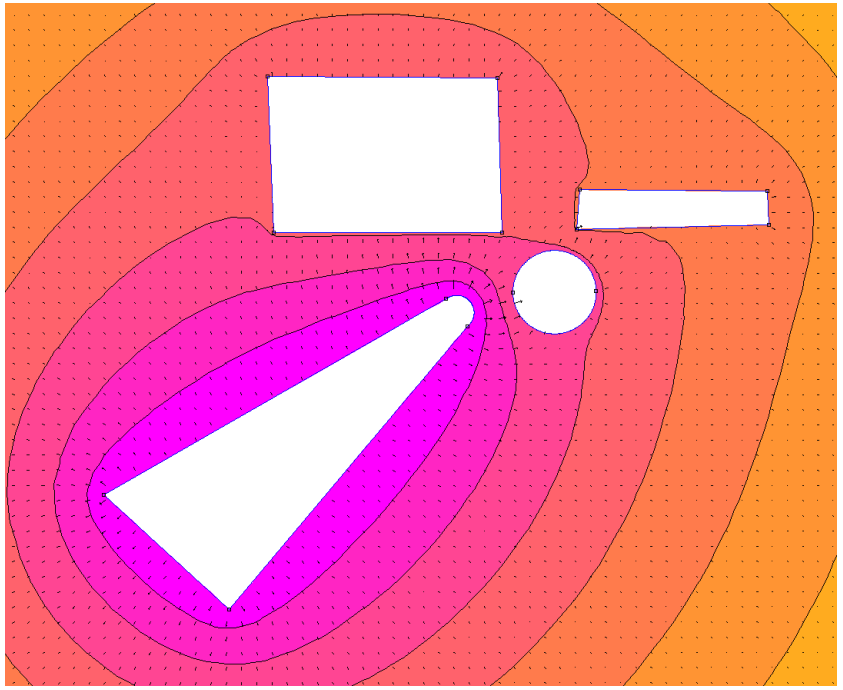
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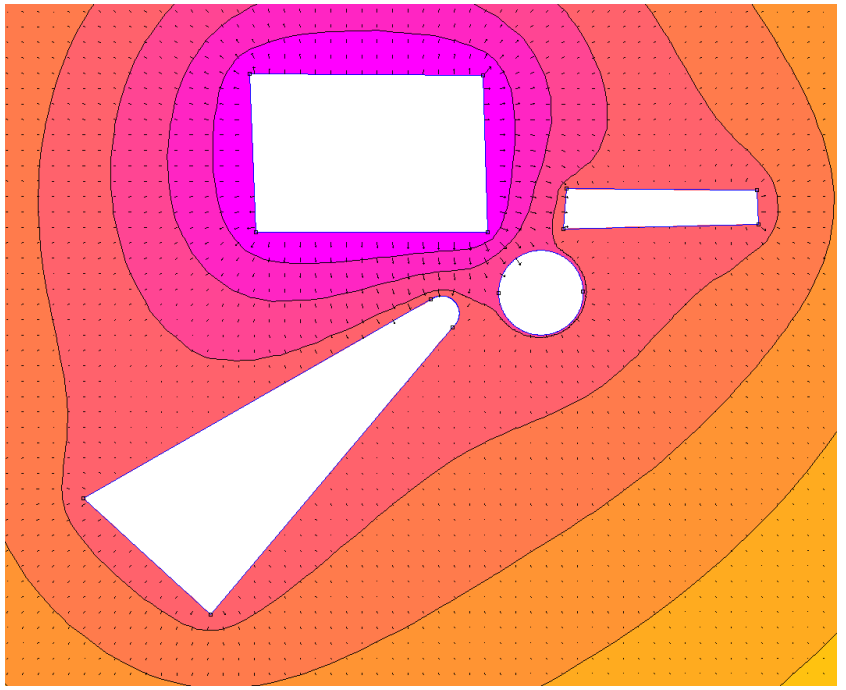
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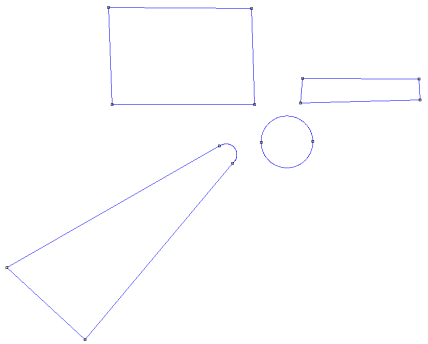


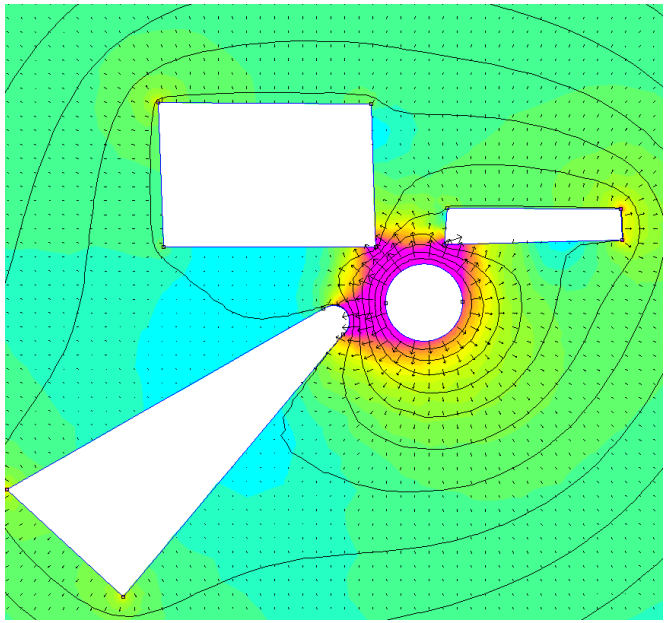


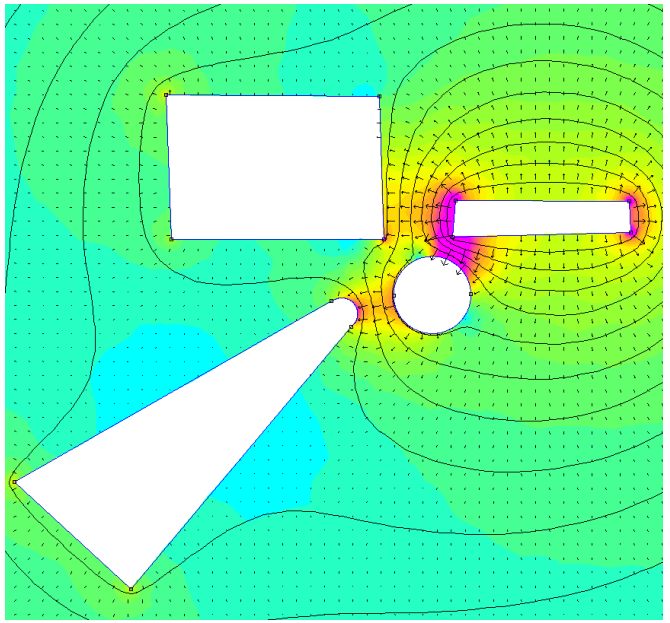


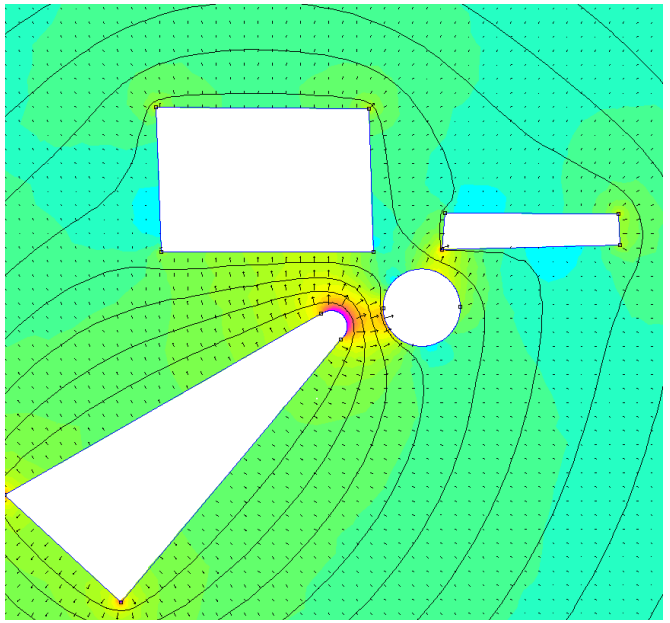


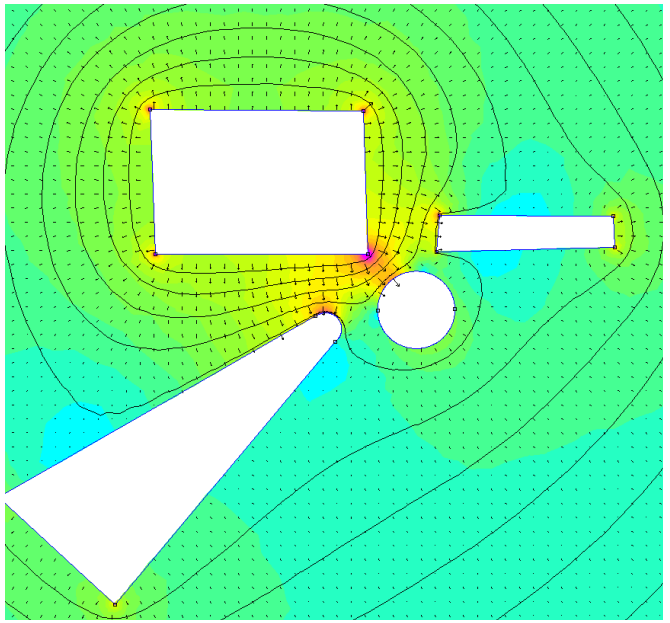


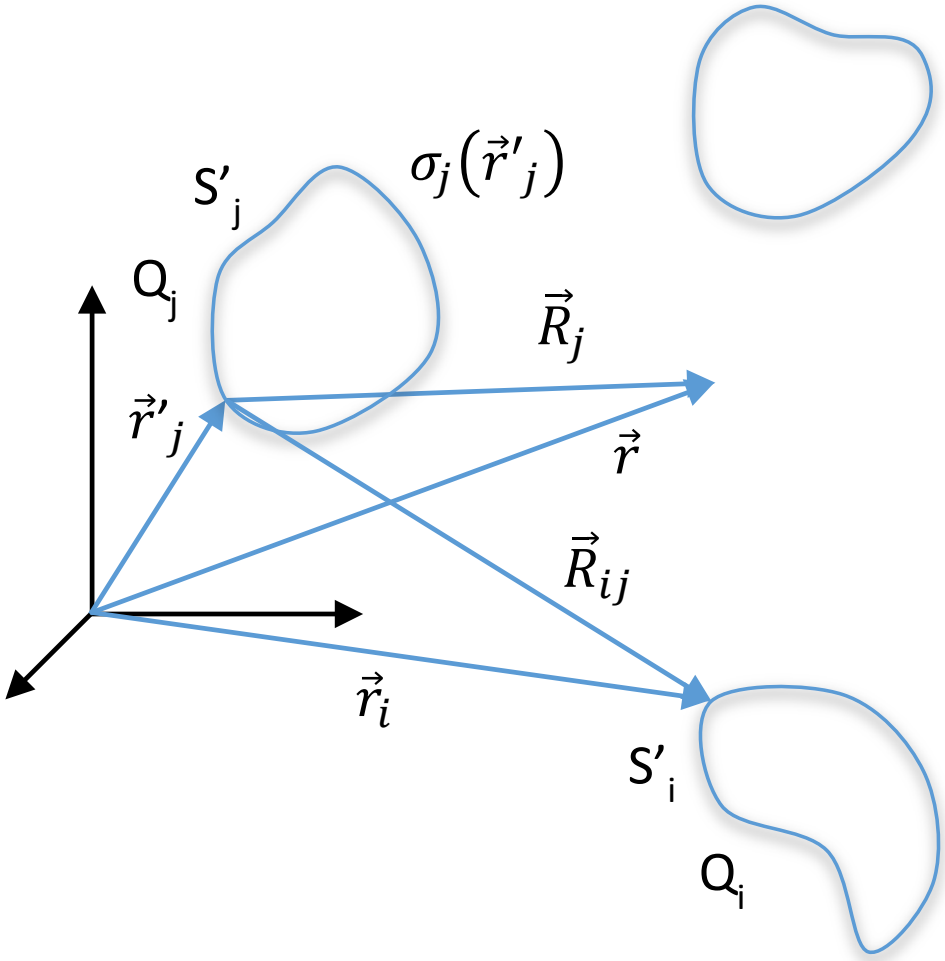











$\sigma_j(\vec{r}'_j)$
S'j
Qj
$\vec{R}_j$
$\vec{r}'_j$
$\vec{r}$
$\vec{R}_{ij}$
$\vec{r}_i$
S'i
Qi